

B.Sc. BOTANY — YEAR I

MAJOR COURSE III

APPLIED BOTANY

UNIT I — Introduction, History and Services of Botany

Detailed Teacher-Style Notes

Definitions • Concepts • Tables • Flow diagrams • Examples • Applications • Exam points

Prepared in the same detailed format as the Major Course II notes.

UNIT I — INTRODUCTION AND APPLICATIONS OF BOTANY

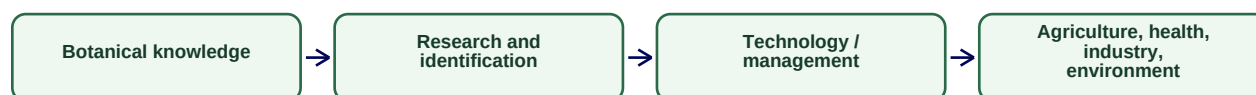
Syllabus topics covered: introduction, objectives and importance of Applied Botany in Indian context; history and evolution of Botany in India; relation of plants with other services; various disciplines of Botany and their applications to human welfare; project on man–plant relations and ancient Indian plants.

1.1 Applied Botany: introduction, objectives and importance

Definition — Applied Botany: The branch of botanical science that uses knowledge of plants, plant processes and plant resources to solve practical problems in agriculture, food, medicine, industry, environment, forestry, biotechnology and human welfare.

- **Basic aim:** convert botanical knowledge into useful products, technologies, management practices and services.
- **Indian context:** Applied Botany is important because India has large agricultural diversity, medicinal plant traditions, forest resources, crop-based livelihoods and major biodiversity-rich regions.
- **Human-welfare link:** plants provide food, fibre, medicines, fuel, timber, ecosystem services and raw materials.

From basic Botany to Applied Botany



Objectives of Applied Botany

- Improve crop productivity and food security.
- Conserve useful plant genetic resources and biodiversity.
- Identify, propagate and sustainably use medicinal and economically important plants.
- Develop plant-based products and industrial raw materials.
- Use plants for environmental monitoring, restoration and pollution control.
- Support plant biotechnology, tissue culture and molecular improvement.

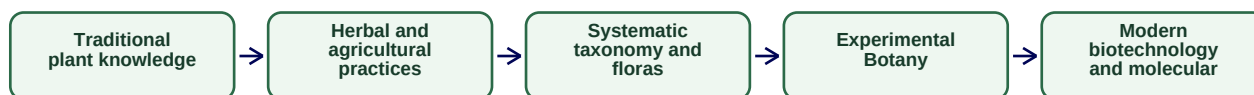
Area	Application of Applied Botany	Example outcome
Agriculture	Plant breeding, crop protection, nutrient management	Higher and more stable yield
Medicine	Identification and cultivation of medicinal plants	Reliable plant raw material
Environment	Phytoremediation and restoration	Cleaner soil/water and habitat recovery
Industry	Fibre, oils, gums, dyes, paper and bioproducts	Plant-based manufacturing
Biotechnology	Tissue culture and molecular methods	Rapid propagation and improved traits

Exam focus: Write the definition first. Then give 5–6 objectives, Indian importance, a table of applications and a short conclusion.

1.2 History and evolution of Botany in India

The history of Botany in India can be studied as a progression from practical knowledge of plants toward systematic taxonomy, plant physiology, ecology, pathology, biotechnology and molecular science. Traditional knowledge emphasized identification and uses of plants, while modern botany introduced standardized observation, experimentation, classification and institutional research.

Broad evolution of botanical knowledge



Phase	Main emphasis	Significance
Traditional knowledge systems	Food, medicine, agriculture and useful plants	Foundation of practical plant knowledge
Classical taxonomy	Naming, identification and classification	Organized plant diversity
Experimental phase	Physiology, genetics and pathology	Explained plant functions
Modern applied phase	Biotechnology, conservation, bioinformatics	Direct technological applications

1.3 Relation of plants with man and other services

Human civilization is deeply dependent on plants. The relationship is not limited to food; it includes ecological regulation, economic activity, health care, culture and livelihoods.

Service	Role of plants
Provisioning	Food, fodder, medicines, timber, fuel, fibre and oils
Regulating	Carbon storage, temperature moderation, water-cycle support and erosion control
Supporting	Primary production, soil formation and habitat creation
Cultural	Religious, aesthetic, traditional and educational value

1.4 Major disciplines of Botany and applications

Discipline	What it studies	Human-welfare application
Taxonomy	Identification and classification	Correct identification of useful and medicinal plants
Anatomy	Internal structure	Wood/fibre studies and diagnostic work
Physiology	Plant functions	Crop and stress management
Genetics	Heredity and variation	Plant breeding and trait improvement
Ecology	Plant–environment relations	Conservation and restoration
Plant pathology	Plant diseases	Disease diagnosis and crop protection
Economic Botany	Useful plants and products	Resource development
Biotechnology	Cells, tissues and genes	Micropropagation and molecular tools

1.5 Project: Man and plant relations; ancient Indian plants

A good project should not merely list plants. Use the structure: **local/ancient plant name** → **scientific identification** → **traditional or modern use** → **part used** → **cultural/economic significance** → **conservation concern**.

Project heading	What to document
Food plants	Staple use, edible part, cultivation and nutrition
Medicinal plants	Botanical name, local name, traditional use and safety of claims
Sacred/cultural plants	Cultural context and conservation value
Fibre/timber plants	Plant part used and economic importance

Exam focus: Unit I high-yield: definition and objectives of Applied Botany; history of Botany in India; branches of Botany; plant services; man–plant relationship.